ECG recording report

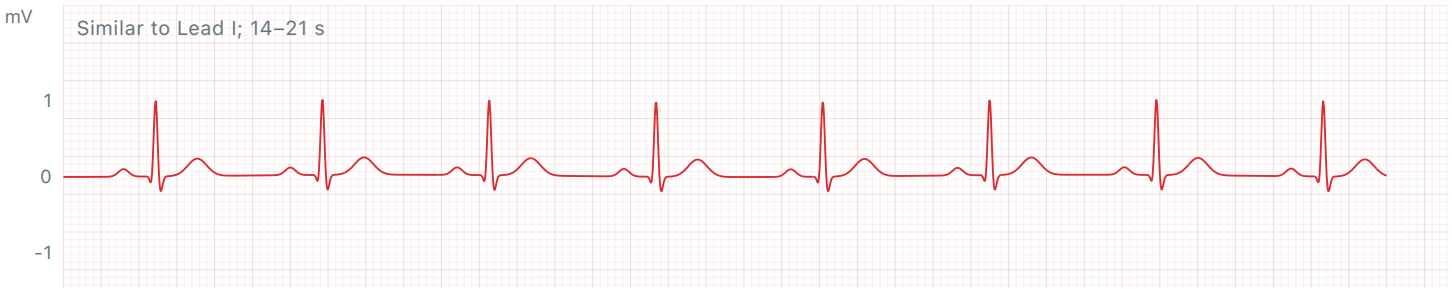
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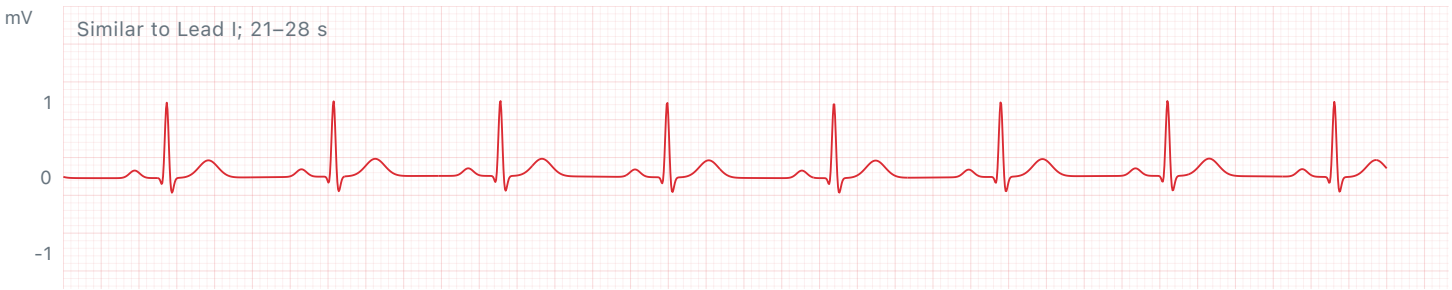
Heart rate	Recording duration	Demo classification
68 bpm	30 s	Sinus rhythm

25 mm/s 10 mm/mV Print at actual size (100%)











Analysis summary

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Average heart rate in this record	Signal coverage	Analyzed duration
68 bpm	100%	30 s

Metric	Result
Data sources	Apple Watch / Synthetic example
R peaks identified	34
Median RR	883 ms
RR standard deviation	1 ms
RMS of successive RR differences	2 ms

RR statistics do not exclude non-sinus beats and are not normal-beat HRV measures.

Premature-beat candidates and rhythm

Rhythm reference score	Premature-beat candidates
100 / 100	Beats: 0

Detected beats: 34; included in analysis: 32.
Flagged beats: 0. Each beat is counted once.

Automatic flags require review against the original ECG. The rhythm score does not assess heart health.

ECG segments

Range is the minimum to maximum in this recording, not a clinical reference interval. The central 80% is P10–P90.
— Unable to measure

P wave

Measurement	Median	Range	P10–P90	Valid / total
P-wave duration	121 ms	109–133 ms	113–133 ms	33 / 34
P-wave amplitude	0.10 mV	0.09–0.10 mV	0.09–0.10 mV	33 / 34

Reasons for missing values: Occurrences: 1. The waveform around this beat is incomplete, or part of the signal is unclear.

PR interval

Measurement	Median	Range	P10–P90	Valid / total
PR interval	188 ms	184–191 ms	184–191 ms	33 / 34

Reasons for missing values: Occurrences: 1. The waveform around this beat is incomplete, or part of the signal is unclear.

ECG segments

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PR segment

Measurement	Median	Range	P10–P90	Valid / total
PR-segment duration	66 ms	59–74 ms	59–73 ms	33 / 34
PR-segment level	0.00 mV	-0.01–0.01 mV	-0.01–0.01 mV	33 / 34

Reasons for missing values: Occurrences: 1. The waveform around this beat is incomplete, or part of the signal is unclear.

QRS complex

Measurement	Median	Range	P10–P90	Valid / total
QRS duration	98 ms	94–98 ms	94–98 ms	33 / 34
Q amplitude (estimated)	-0.07 mV	-0.08–-0.06 mV	-0.07–-0.06 mV	33 / 34
Q duration (estimated)	20 ms	16–20 ms	16–20 ms	33 / 34
Main R-peak amplitude	0.94 mV	0.92–0.95 mV	0.92–0.95 mV	33 / 34
QRS start to R peak	43 ms	39–43 ms	39–43 ms	33 / 34
S amplitude (estimated)	-0.18 mV	-0.19–-0.17 mV	-0.19–-0.17 mV	33 / 34
S duration (estimated)	31 ms	27–35 ms	28–35 ms	33 / 34

Reasons for missing values: Occurrences: 1. The waveform around this beat is incomplete, or part of the signal is unclear.

ST segment / J point

Measurement	Median	Range	P10–P90	Valid / total
ST-segment duration	59 ms	47–66 ms	47–66 ms	33 / 34
J-point level	-0.01 mV	-0.02–0.01 mV	-0.02–0.00 mV	33 / 34
Level 60 ms after J	0.02 mV	0.01–0.02 mV	0.01–0.02 mV	9 / 34
Level 80 ms after J	—	—	—	0 / 34
ST voltage change rate	—	—	—	0 / 34

Reasons for missing values: Occurrences: 33. The T-wave start is unclear, so the ST measurement window cannot be determined.; Occurrences: 33. This position is too close to the T wave to measure ST separately.; Occurrences: 1. The waveform around this beat is incomplete, or part of the signal is unclear.

ECG segments

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T wave

Measurement	Median	Range	P10–P90	Valid / total
T-wave duration	219 ms	195–254 ms	199–246 ms	33 / 34
T-wave amplitude	0.23 mV	0.22–0.24 mV	0.22–0.24 mV	33 / 34
T-wave duration ratio	0.97 ×	0.79–1.07 ×	0.90–1.04 ×	33 / 34

Reasons for missing values: Occurrences: 1. The waveform around this beat is incomplete, or part of the signal is unclear.

QT interval

Measurement	Median	Range	P10–P90	Valid / total
QT interval	371 ms	359–395 ms	359–394 ms	33 / 34

Reasons for missing values: Occurrences: 1. The waveform around this beat is incomplete, or part of the signal is unclear.

QTc (corrected QT)

Measurement	Median	Range	P10–P90	Valid / total
QTc (Fridericia)	387 ms	375–412 ms	375–407 ms	31 / 34
QTc (Bazett)	395 ms	382–421 ms	382–416 ms	31 / 34

Reasons for missing values: Occurrences: 2. The QT end is unclear, or nearby RR intervals do not meet the correction conditions. See Measurement method for details.; Occurrences: 1. The waveform around this beat is incomplete, or part of the signal is unclear.

TP segment

Measurement	Median	Range	P10–P90	Valid / total
TP-segment duration	324 ms	297–340 ms	301–336 ms	32 / 34

Reasons for missing values: Occurrences: 1. The waveform around this beat is incomplete, or part of the signal is unclear.; Occurrences: 1. This T-wave end or the next P-wave start is unclear, or there may be a signal gap between them.

RR / PP intervals

Measurement	Median	Range	P10–P90	Valid / total
RR interval	883 ms	879–883 ms	880–883 ms	33 / 34
PP interval	883 ms	879–883 ms	883–883 ms	32 / 34

Reasons for missing values: Occurrences: 1. The previous beat is missing, the P wave is unclear, or the signal has a gap between beats.; Occurrences: 1. The waveform around this beat is incomplete, or part of the signal is unclear.

Measurement method

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These values are estimated automatically by the app. Accuracy has not been validated on real Apple Watch data, and the values cannot determine whether an ECG is normal.

Metric	Result
Data sources	Apple Watch
Sampling rate	256 Hz
Analysis version	0.4.0-research

Detection and scoring rules

Rhythm reference score = 100 × (1 – flagged beats ÷ reviewable beats), rounded. Flags include premature-beat candidates, marked interval changes or local waveform changes. The pause after a candidate is not counted again. This describes the share of beats not flagged by these rules in this recording. It is not a heart-health score, and 100 does not mean the ECG is normal.

At least 20 seconds of clear signal, 80% signal coverage, 15 reviewable beats and 8 stable reference beats are required. At least 80% of non-edge beats must be reviewable. Poor signal does not reduce the score; missing PR or QT values and higher or lower HRV do not add or subtract points.

Compare with nearby beats: the beat must arrive over 15% and at least 80 ms early, with a longer following interval or a wider, changed QRS. Wide candidates require a local width of at least 120 ms, over 20% wider than the reference, and template correlation below 0.90. Narrow candidates require a width under 120 ms, correlation of at least 0.90 and a longer following interval. These research rules cannot by themselves confirm atrial or ventricular premature beats.

Rule version: rhythm-review-1

P wave

The start, end and peak of the P wave are estimated. If it is too small, its boundaries are unclear or it overlaps nearby waves, the affected values are left blank.

PR interval

Uses the P-wave start and QRS start from the same beat. If either is unclear, no PR interval is calculated. PR includes the P wave itself and the following PR segment.

PR segment

Measured from the P-wave end to QRS start. The level is the mean voltage of this segment relative to the local TP baseline. Positive values are above the baseline; negative values are below it.

QRS complex

QRS start and end are estimated first. Opposite-direction waves before and after the main peak are candidates for Q and S. Values retain their original polarity. If the main peak points down, Q, R and S labels need manual review. A single lead cannot assess the electrical axis or R-wave progression across chest leads.

Measurement method

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ST segment / J point

The QRS end estimates the J point. Voltage is measured 60 and 80 ms later. The change rate comes from the voltage difference between these positions: positive means upward and negative downward. Positions too close to the T wave are not used for ST measurements.

T wave

The T-wave start, peak and end are found separately. The duration ratio is start-to-peak time divided by peak-to-end time. It is near 1 when the two parts have similar lengths. Multiple peaks require checking the original trace.

QT interval

Each beat is measured separately, from its QRS start to its T-wave end, before results are summarized. If the T-wave end is unclear, QT is not calculated.

QTc (corrected QT)

Uses QT from the same beat and the immediately preceding RR, both in seconds: Fridericia = QT / RR^(1/3); Bazett = QT / √RR. The algorithm requires continuous neighboring RR intervals, changes no greater than 10%, and intervals of 500–1500 ms. These are algorithm conditions, not a normal range.

TP segment

Measured from this T-wave end to the next P-wave start. No value is produced if either boundary is unclear, the signal has a gap, or the interval is shorter than 40 ms.

RR / PP intervals

RR and PP are calculated from their own consecutive peaks. They do not substitute for one another, and intervals are not joined across signal gaps.

Recording notes

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Duration: 2 min
Caffeine

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